2024-01-29

Name at least three metrics (one of them must be graphical) that are used in the verification scenarios and another three (one again graphical) for identification scenarios.

Verification: FMR, FNMR in DET

Identification: precision, recall in CMC

What are the minutiae points? Which types do we know? How do we determine the direction of each minutiae point?

Minutiae are **abstract representations** of the **ridge skeleton** and **capture** a **lot** of **discriminative information**.

Minutiae are locations where a ridge: **emerges**, **ends**, **splits** or **merges**.

If it is an end, we take a fixed size of pixel that we move back to and calculate an angle to that point.

If it is a bifurcation, we take a fixed size to both pixels and take a mean angle.

Which metric do we normally use for matching in the iris recognition pipeline? What can you say about the imposter and genuine examples distributions in this context?

We use Hamming distance (normalized or weighted).
Normalized Hamming distance: fraction of bits that differ using masks (1 iris, 0 occlusion).
- 0.0 – no difference
- 1.0 – totally different

The impostor distribution tends to have a higher HD than the genuine distribution.
With shifts (that we use for accounting for the head tilt), the distribution stays somewhat similar with the impostor peaks being higher with correspondence to the number of shifts.

What do we normally do in the iris recognition pipeline to account for the varying head tilt? How is that implemented? What is the main advantage and the main disadvantage of this process?

We rotate codes. Metric is computed for a range of circular shifts and the minimum is kept.

We can use bit shifts to **correct the authentic distribution**, but it **increases time for matching** and **changes the impostor distribution** to a more focused frequency.

How do we calculate the integral image? Where is it used? What is a weak and what is a strong classifier in this context?

We calculate the integral image with the sum of all pixel intensities above and to the left of a specific (x, y) pixel. It is used as a computation reduction in the Viola-Jones algorithm. A weak classifier is a classifier that only needs to exceed chance performance. A strong classifier is a weighted linear combination of the best weak classifiers.

Describe the alignment by geometric warping. Which method is better addressing the alignment than this one?

The method aligns the face transforming the image. It takes the eyes positions as an input together with the horizontal and vertical offsets and the image size as the output parameters.

The output is a scaled, rotated and cropped image.

A better method for general alignment is “Landmark Detection” as it is not limited to faces.

What is a jet in the context of face recognition? In which method do we use them? How do we construct them and use them in matching?

A jet is a set of Gobel coeficients of the local texture from a central fiducial point. A jet characterizes the local texture information.

We use them in Elastic Bunch Graph Matching – Face Bunch Graph.

We construct them by calculating the Gobel coeficients by convolving the image with a complex 2D Gabor filter.

Take the mean model Gabor jets at each bunch and find approximate face position. We compute the similarity as the average similarity between the jets of the image graph from the probe and the model graph. Lastly, we take best similarity score and apply a threshold.

Describe the properties of a multimodal system.

- It combines different types of modalities together.
- Traits are expected to be uncorrelated, which leads to greater improvement in performance compared to other types of multibiometric systems.
- Higher cost.
- There are diminishing returns after strong biometric traits have been fused.

Which combination rules do we know in transformation-based score fusion? Which assumptions do they have to follow? What is the key difference between score-level fusion in verification and identification mode?

Combination rules: product, sum, max, min and median.

Assumptions: match scores of different matchers are statistically independent, prior probabilities of the genuine and impostor classes are equal.

In verification system the score data is in the form of a vector, while the score data in an identification system is in the form of a NxM matrix.

Illustrate and describe the use of HOGs for matching faces.

Image descriptors invariant to 2D rotation. The face image (based on landmarks) is divided into cells. For each cell, we compute a histogram of gradient directions. Then, histograms of all cells are concatenated into a single feature vector. At the end we compare these vectors for a match.

2023-01-18

What is the difference between authentication types and recognition types?

Authentication types: Pair matching, Verification, Identification, Negative authentication, Deduplication

Recognition types: Positive recognition, Negative recognition

Authentication types are methods used to confirm or verify a subject's identity, while recognition types refer to if a subject is known or unknown to the system.

What is wrong with the accuracy as a detection metric (explain through an example)?

Can you name and explain a more suitable one?

A sliding window face detector is used on an image with 11 faces and 1000 negatives. The majority class identifier, which predicts "no-face" would be 99% accurate because of the high dataset imbalance. But the actual face classifier has an accuracy of about 1%.

More suitable could be precision, recall, F1-score.

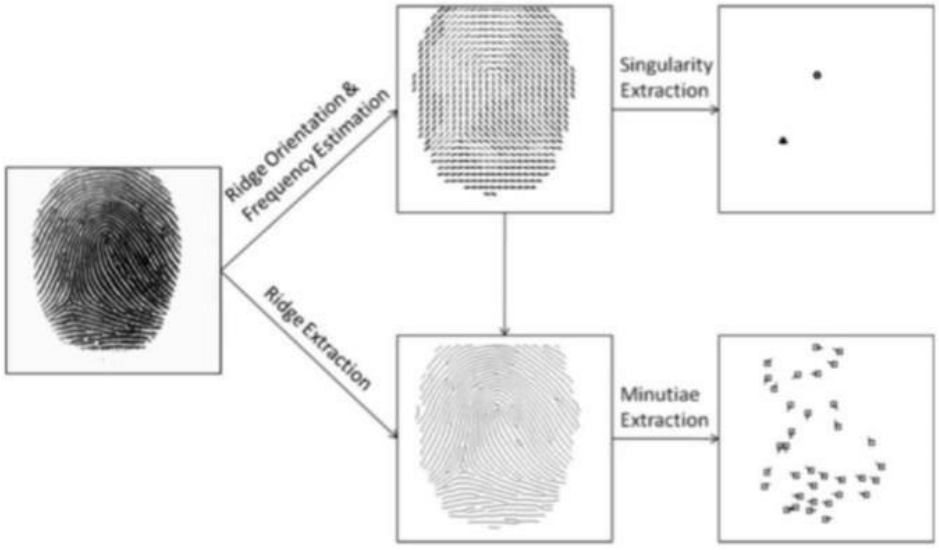
Precision is TP (correctly identified faces) among all positives predicted, while accuracy is TP + TN divided by all TP, TN, FP, FN.

Explain the information given on a tenprint card.

Two rows of 5 for each finger on both hands. (Row 1 – right, Row 2 – left) -> ROLLED

The last row has 4 columns. The first column is four left fingers taken at the same time. Then the left thumb. Then the right thumb. Then four right fingers taken at the same time. -> PLAIN

Draw a sequence of steps that result in fingerprint feature extraction at level 1 and 2. Explain briefly each step.



What iris resolution do most iris recognition systems expect for successful processing?

Which is the fundamental operator for comparison of iris codes?

The resolution of **100 to 200** pixels.

Operator XOR (Hamming distance).

Why the pupil size varies (there are plenty of reasons)? What did Daugman state in his patent about the influence of dilation? How did he solve the problem of different pupil sizes?

The main concerns are dilation and contraction.

He stated that a pupil sample lasts a lifetime, as the iris texture is essentially immutable.

Daugman's **rubber sheet model**. Linearly stretching or compressing the imaged iris to a standard size frame.

Which step of the Viola-Jones face detector can be regarded as a brute force approach and why? Why does this method have problems with inverted faces?

The first step could be regarded as a brute force approach as it scans through the input image with windows of different sizes without any initial assumptions.

It has a problem with inverted faces because it uses Haar-like features, which are specific patterns of contrast between light and dark regions. Also, because the classifiers are trained with upright faces and non-face objects/images.

Explain the basic concept of the PCA. How do we reduce the dimensionality of the space with it?

PCA – Principal Component Analysis: it's a kind of dimension reduction, which maximizes the variance within the same class or all the data in question.

Uses the training data to learn the subspace that accounts for as much variability in the training data as possible by performing an Eigen value decomposition of the covariance matrix of the data.

A lower dimensional subspace, which is spanned by d' (d' < d) Eigen vectors from E corresponding to the largest Eigen values. It accounts for most of the variability in the data by only a few Eigen vectors with the largest values in descending order.

TLDR: We perform the Eigen value decomposition of the covariance matrix of the data and pick the K largest Eigen vectors, which account for a big part of the variance in the data, while reducing the dimensions.

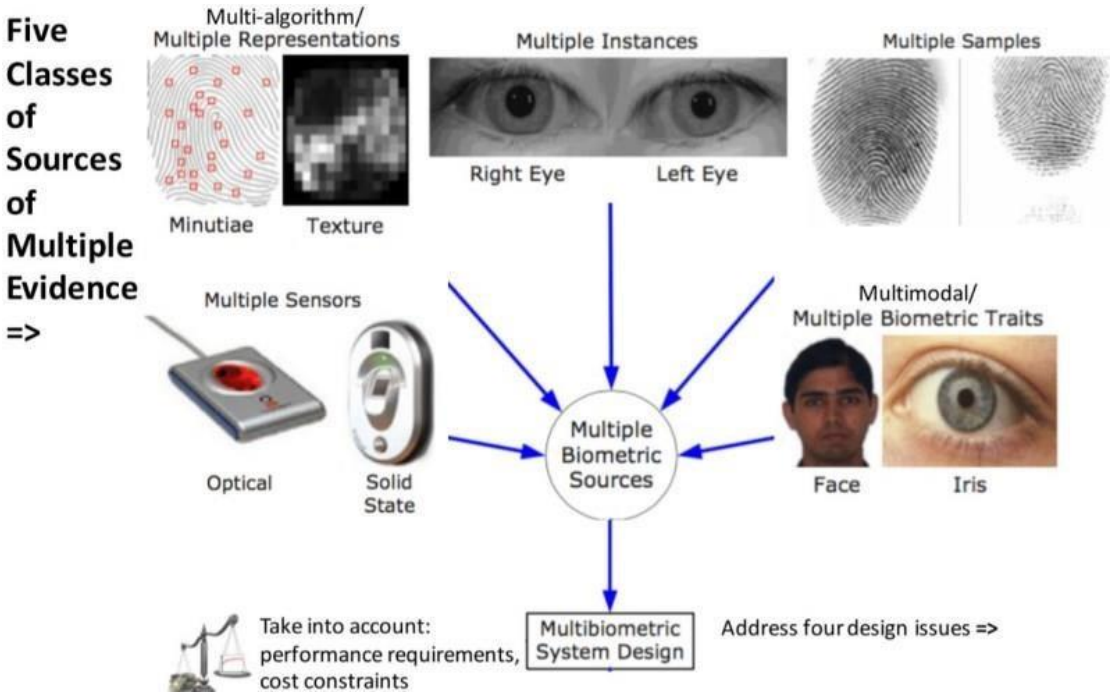
Explain how MLBP method works.

It's a simple texture operator, which features are obtained from a neighborhood region.

The method uses two parameters: number of sampling points and a radius. If a sampling point is off the pixel grid, bilinear interpolation of pixel values is used to get the intensity value of the sampling point. Each pixel is encoded by comparing the neighboring intensity values to the central pixel and get the result as the binary code, which can be further converted into a decimal number.

>> After the encoding, the image is divided into several windows and the histogram in each window is computed. The number of bins in the histogram is 8 (LBP) and 2^P (MLBP). We get the global feature vector by concatenating all histograms and normalizing the final vector.

Draw and briefly explain the taxonomy of five classes of biometrics sources of multiple evidence.



Multiple Instances – Example: 3 different fingers, 2 different eyes

Multiple Samples – Example: same finger 3 times

2022-01-26

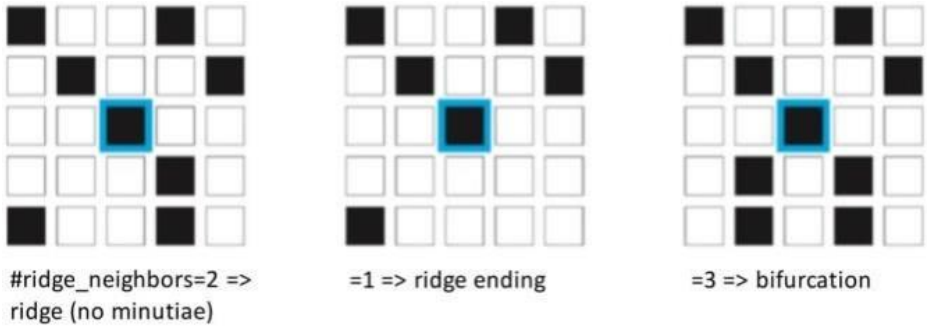
Explain all three levels of features for face recognition.

L1 – geometry of the face and global skin color (short round face / elongated thin face, male / female, different races)

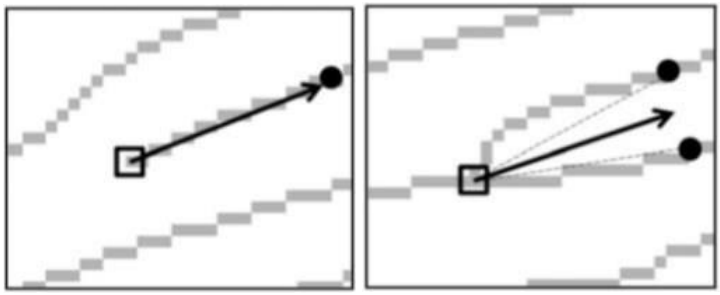
L2 – structure of the face components (eyes), the relationship between facial components, precise shape of the face

L3 – micro level features on the face like scars, freckles, moles, …

We skeletonized the binary image of a fingerprint. How do we now extract the minutiae information (including direction)? Sketch and explain.



One neighbor -> Ridge ending; Two neighbors -> Ridge; Three neighbors -> Bifurcation



If it is the ending -> Trace the ridge to a fixed distance (10) – get direction

If it is bifurcation -> Trace the ridges to a fixed distance for both points and calculate the mean

What kind of pattern types can we observe on rolled fingerprint images? What can we conclude based on the number of singular points on them?

We can observe all pattern types on the rolled fingerprint images, because of a more complete impression. If there are **0** singular points it is a “plain arch”, if it has **2** singular points it is either a “tented arch”, “left loop” or a “right loop” and if it has **4** singular points it is either a “whorl” or a “twin loop”.

Explain alignment by geometric warping. Which method gives better results in a more general alignment context and why?

The method aligns the face transforming the image. It takes the eyes positions as an input together with the horizontal and vertical offsets and the image size as the output parameters.

The output is a scaled, rotated and cropped image.

A better method for general alignment is "Landmark Detection" as it is not limited to faces.

Why is the normalization of iris images needed in a standard iris recognition pipeline? How do we do it?

Normalization addresses the problem of **variations in pupil size** across different images, sensors, …

Daugman's **rubber sheet model**, with which we linearly stretch or compress the iris to a standard size frame.

How can we test our trained model using cross validation? Why do we use cross validation?

We can test our model by most commonly using the "**K-fold**" method. We choose a K number, which gives us N/K training sessions of N-K samples.

In practice let's say we use 10-fold -> We split the dataset to N/10 sets, one set is the validation set others are the training set. We repeat for each set.

We use **cross validation** to **prevent overfiting** and **improve on generalization**.

What is an integral image? Where did we use it? How did we use it? Why did we use it?

An integral image is a representation of the original image where each point is the sum of all pixel intensities above and to the left of the (x, y) pixel.

We used it in the Viola-Jones algorithm.

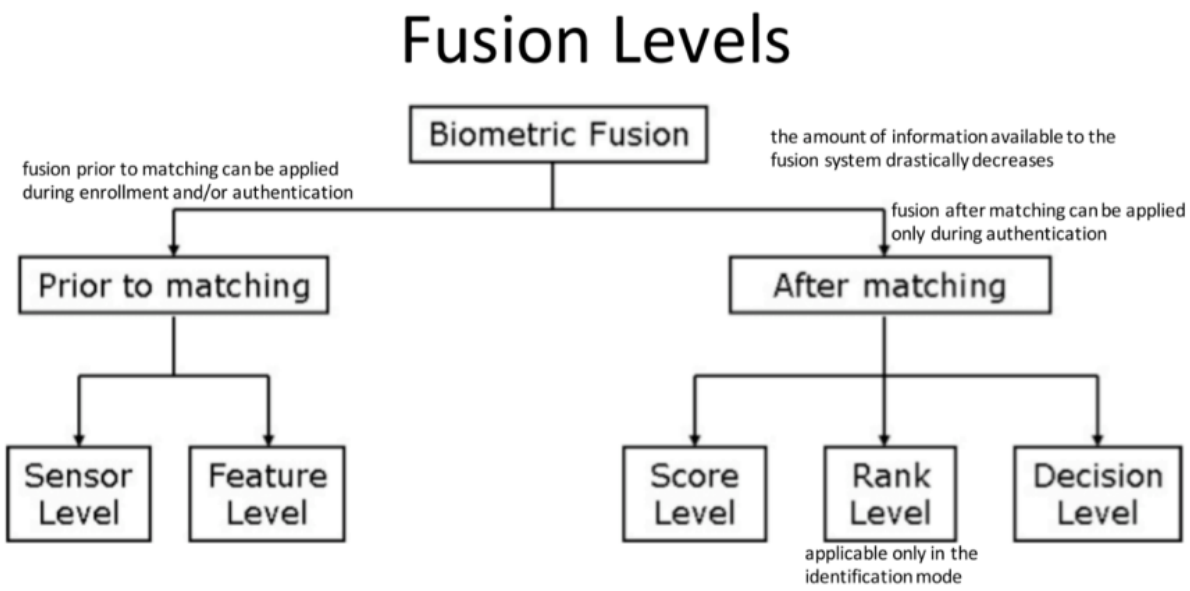
We used it by calculating it based on the input image with a formula: **S(x, y) = SUM I(x', y')**

We used it to reduce the computational burden, because of the number of Haar-like features that can be derived from a single window.

Which modality out of three in-depth discussed at the course gives most interpretable information to a human for determining the gender of a subject based on extracted features? Explain why?

The most interpretable information about gender is given by the face modality, because as compared to fingerprint or iris, which can offer some insight into gender, the face modality has immediate facial features and gender cues such as facial hair, jawline and the distribution of facial fat.

Sketch the taxonomy of fusion levels. Which of them are applicable in a verification scenario?



Verification: because “verification == authentication” => all levels but “Rank level” can be used

Place the formula D = [(F1+F2+F3): F2/(F1+F3)] into the context of the course content and explain it.

Iris quality assessment. Used to compute quality descriptor values at analyzing the Fourier Spectra of Local iris regions. Variables are – power of the low, middle and high frequency components.

2022-02-09

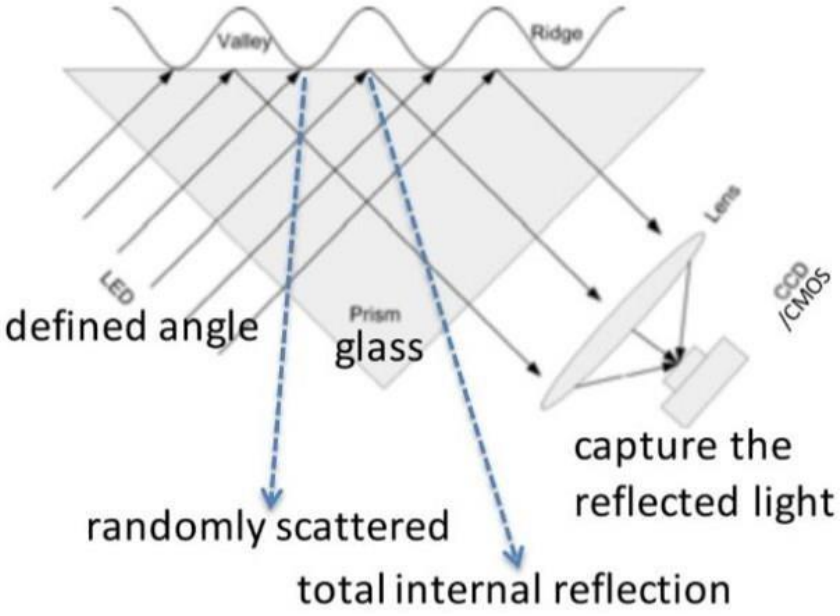
Describe the main differences between the three levels of features for fingerprint recognition.

L1 – orientation field or ridge flow and singular points (loop, delta) (**250** ppi)

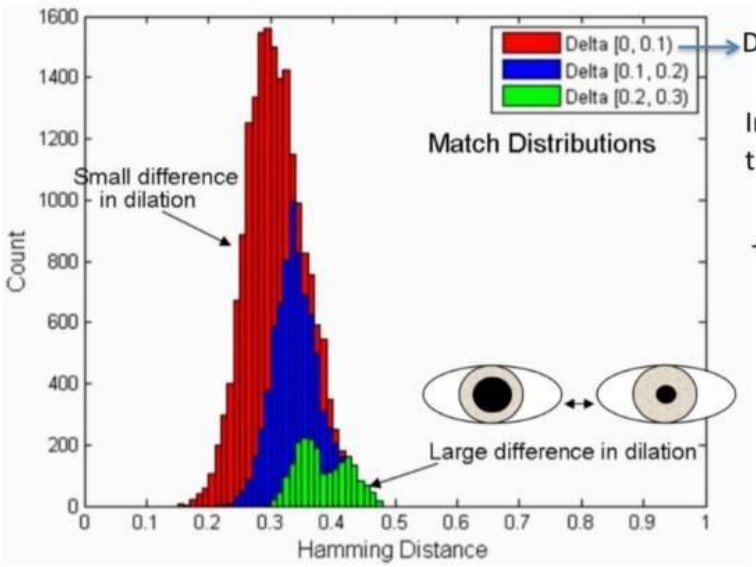
L2 – ridge skeleton and minutiae (locations where ridge emerges, ends, splits or merges) (**500** ppi)

L3 – ridge contour, pore, dot (**1000** ppi)

Sketch the design of an optical FTIR-based sensor and explain on it how the image is generated.



What conclusions did we obtain from iris recognition system when we experimented with images of the same person but with different pupil dilations (in terms of authentic and impostor distributions)? Sketch and explain.

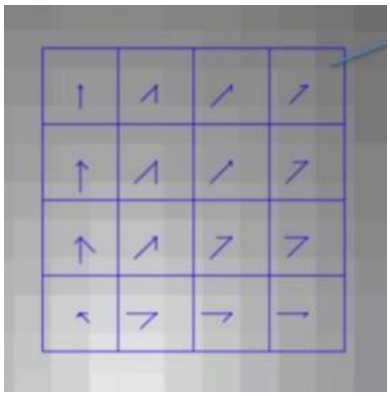


Difference in dilation ratio

Increased difference in dilation **shifts** the authentic distribution

+No change in impostor distribution

Sketch and explain HOG image descriptors. Where did we use them? What is the difference between sparse and dense arrangements?



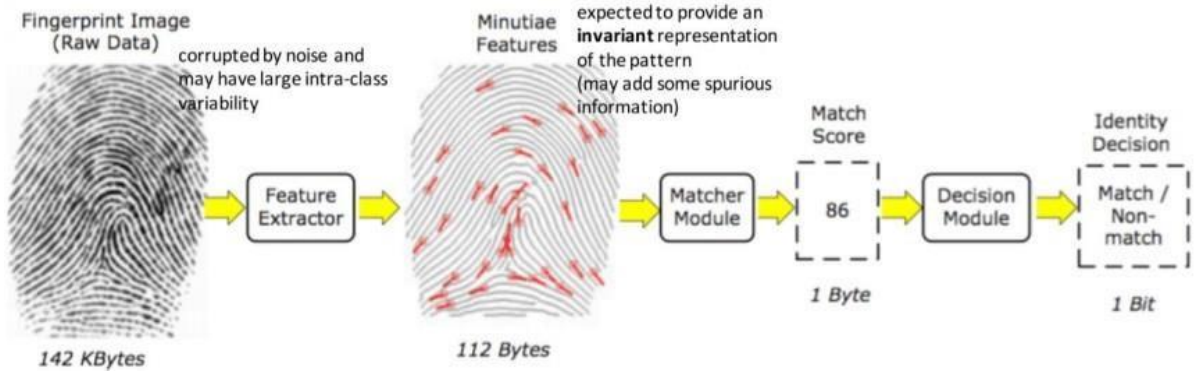
One square is 8 pixels -> 8x8 patch -> 8x8 HOG descriptor. Each cell shows the orientation of the gradients. HOG counts occurrences of edge orientations.

We used HOG for **face recognition**.

In **sparse arrangement**, HOG descriptors are computed at key locations and typically faster to compute.

While in a **dense arrangement**, HOG descriptors are computed over a grid of uniformly spaced cells.

We said that the amount of information available for fusion gets reduced as one progresses along the various processing modules of a biometric system. Select one modality and sketch and explain this statement on it.



[Input Image] → [Sensor Level] → [Feature Extraction] → [Matching] → [Decision]

If using multiple sensors the fusion can happen, but it is still rich.

Features like edges, contours, landmarks are extracted. Ex: background elements are discarded.

Only extracted features are used to get similarity score. Much less information.

It is a binary output – match/no-match.

How can we detect the presence of printed contact lenses in ocular images? Can the printing process affect the detection – and if yes, how?

A printed iris typically exhibits some artifacts that can be detected by analyzing the 2D Fourier spectrum of the iris image. Other approaches are texture analysis and no iris distortions when the pupil changes in size.

Yes, it can intentionally create a false non-match.

What is the main difference between Eigenfaces and Fisherfaces? Write the algorithm (in pseudo-code) that gives Eigenfaces as an output.

Eigenfaces attempt to maximize the scatter of the training images in faces space (**PCA method**)

Fisherfaces attempt to maximize the between class scatter, while minimizing the within class scatter (**LDA method**)

- Compute the average of the training set: $\mu = (1/N) \Sigma xi$
- Data matrix: $X = [(x1 - \mu), (x2 - \mu), ..., (xN - \mu)]^{i=1}$
- Covariance matrix: $C = X * X^T$
- Find eigenvalues and eigenvectors of C in E.
- Any data can be represented as a weighted sum: $\omega = E^T * x$ (subspace spanned by chosen eigenvectors)

Where did we use Borda Count Method and Logistic Regression Method? How do we get the first one from the second one?

Borda Count Method and Logistic Regression method are under **Rank-level fusion**.

Borda Count Method uses the **sum of the ranks** assigned by the **individual matchers**, while **Logistic Regression Method** is a generalization of the Borda Count method where a **weighted sum** of the **individual ranks** is **calculated**.

Name and explain the properties of curves giving FNMR and FMR dependence. Additionally, let's say that we are observing two curves that are intersecting at one point: how can we argument which one (which system) is better?

Two forms: Receiver Operating Characteristic curve and Detection Error Tradeoff curve

ROC – plots TPR against FPR at various thresholds. Closer to upper left, better performance.

DET – plots FNMR against FMR at various thresholds. Closer to bottom left, better performance.

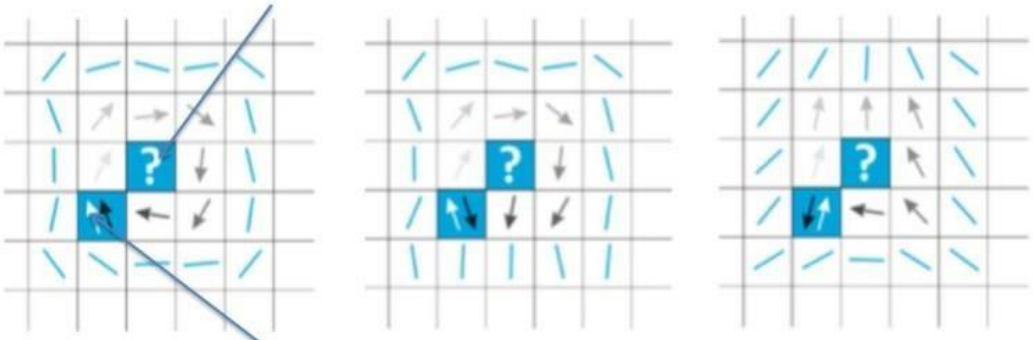
When two curves intersect, it indicates that each system performs better in different operating regions.

Which is better depends on the application:

- If **Security** is important, prefer the system that provides a lower FMR before the intersection.
- If **User experience** is important, prefer the system that provides a lower FNMR after the intersection.

Sketch and describe the method for extraction of singularity points.

calculating the Poincaré Index



$p = 360^\circ$ (whorl)
(white and black arrow are in the same direction)

$p = 180^\circ$ (loop)

$p = -180^\circ$ (delta)

1. Select a random orientation for **one** example element (white) of the directional image
2. For the remaining elements select the **closest** orientation to the neighboring element
3. Travel along the closed curve and sum up the rotations

if 0° => no singular point

2020-01-29

Describe decision-level fusion. In this context describe majority voting method.

It's an **abstract level**, which only gives us a **1-bit** – final level **recognition decision**.

The **input sample** is assigned to the class on which **a majority of the matchers agree** (half). It assumes that **all** the **matchers** perform **equally well**.

Describe the properties of multimodal biometric systems.

- Traits are expected to be uncorrelated: which leads to better performance.
- Higher cost
- Diminishing return after strong biometric traits have been fused: in some cases, there may even be performance degradation, if lower accuracy traits are added.

What can you say about the analysis of the discriminative information contained in each layer of a CNN?

Higher levels generally produce **more discriminative features** (as feature hierarchies become deeper, they learn increasingly powerful features).

What is the main difference between PCA and LDA? How do Fisherfaces and Eigenfaces correlate to them? What should we do if the number of training samples (N) is smaller than the dimensionality of the data (d)?

PCA maximizes the variance **inside the class** and **minimizes** the variance **between the classes**, while **LDA maximizes** the variance **between the classes** and **minimizes** the variance **inside the class**.

If "N" is smaller than "d": apply PCA to the training samples to reduce the data dimensionality and then apply LDA to the transformed data having lower dimensionality.

Describe alignment by geometric warping. Any problem with this method? If yes, what kind of a problem and how can we mitigate it?

Geometric warping performs **in-plane rotation** based on the **eyes to align the face**. The problem with this method is that it **only rotates** the image in a **single plane** and limits the ability to align faces. We can mitigate the problem by using **Landmark Detection** method.

How and why do we use cascades in the Viola-Jones method? Which are the two most effective Haar-like filters? In which stage of the cascade would you use them?

In Viola-Jones, cascades are used to **reduce computation** by eliminating as many false images as soon as possible and connect lower classifiers into a stronger one. The first is **2-rectangle** filter "**black top, white bottom**" (darker eye region, lighter cheek region). The second is **3-rectangle** "**black left, white middle, black right**" (brighter nose bridge compared to the eye regions). We would use them in the **first stages of the cascade**.

Describe and link the following terms together: singular points, pattern type, minutiae, resolution.

L1 fingerprint features are called **singular points**, which are abstract representations of the **pattern types**, which are decided based on the number of loops and deltas. However, if we have a higher **resolution** sensor, we can analyze the L2 features which are called **minutiae**.

What kind of image imperfections can we find by analyzing the Fourier spectra of local iris regions? Describe the properties of the Fourier spectra for each of them.

- **Out of focus blur**: Fourier spectrum is supposed to be largely dominated by the low frequency components.
- **Motion blur**: Lacks middle and high frequency components and has frequency distribution similar to that of an out-of-focus image.
- **Occlusion due to the eyelashes, eyelids**: Contains significant middle and high frequency components.

Why do we need to normalize the iris image? Describe the normalization process. Why do we need the iris mask for?

Addresses the problem of **variations in pupil size** across multiple images, subjects, sensors, …

Main concerns are **pupil dilation** and **contraction**. We use the Daugman's **rubber sheet model**, which is linearly stretching or compressing the imaged iris to fit a size frame.

We use a mask to **mask out the noise in the image** (eyelids, eyelashes), because we don't want them to **impact** the **feature extraction**.

How do we extract minutiae from (one-pixel wide) ridges? How do we determine their directions?

We look at a point and its immediate neighbors. If it has two ridge points then it is just a ridge – no minutiae. If it has one ridge point then it is a minutiae (ending). If it has three ridge points that it is a bifurcation.

For an ending, we determine the direction by taking a fixed size and move this much back. We then calculate the angle. For a bifurcation, we move a fixed size to both sides and take the mean angle as direction.

2018-01-31

Which of the statements are scientific facts?

a) **The physical trait itself is unique.**
b) **The physical trait itself is permanent.**
c) **Biometric systems rely only on the digital measurements and not on the real physical traits.**
d) **Sensing introduces variations in the samples of the same biometric trait of a user obtained over a period of time.**
e) **If two feature sets are identical, it may be a strong indication that the biometric data actually comes from an adversary who is replaying the data recorded at an earlier time.**

A, B – NOT A FACT C, D, E – TRUE FACT

Name problems pertaining to higher error rates (FNMR).

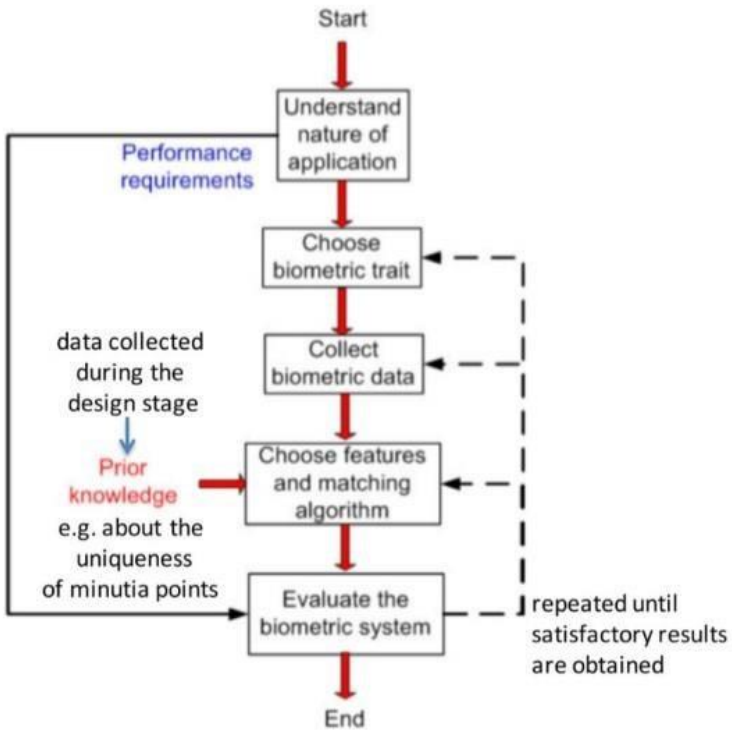
Large throughput volume is a problem.

A system where each person is 1:1 verified at an airport kiosk with 5000 people per hour (14h/day) requesting access.

Explain the ROC curve and what it reveals if we compare two systems.

Receiver Operating Characteristic Curve. It **simplifies** the **comparison** of **two** systems with **different** thresholds.

Name and sketch general steps in the design cycle of a biometric system.



Which are the levels of features in fingerprints; name them by features? Define the meaning of singular points. What kind of singular point types are we looking for within the image?

L1 – orientation field and ridge flow and singular points

L2 – ridge skeleton and minutiae

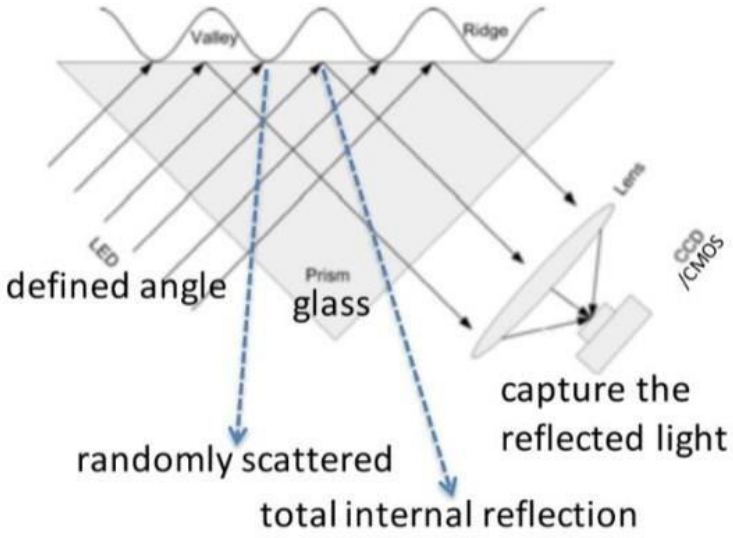
L3 – ridge contour, pore, dot

Singular points are **locations** where the **ridge orientations change abruptly**.

Two basic types are **LOOP** and **DELTA**.

Name fingerprint scan sensors technology. Describe one of them in detail.

Optical FTIR-based sensor, Capacitance-based sensor, Ultrasound reflection-based sensor, Piezoelectric effect-based sensor, Temperature differential-based sensor



Dificult to have this arrangement in a compact form; focal length of small lenses can be very large.

Image distortions are possible when the reflected light is not focused properly.

Sketch basic taxonomy of fingerprint quality evaluation algorithms. Explain briefly how NFIQ (NIST Fingerprint Image Quality) algorithm works.

QUALITY EVALUATION => LOCAL PROPERTIES || GLOBAL PROPERTIES

LOCAL:

- Local ridge orientation
- Gabor filters
- Pixel intensity
- Power spectrum

GLOBAL:

- Field direction continuity
- Uniformity of the frequency field
- Features counting

NFIQ method combines **local** and **global** properties. For each feature (minutiae) calculate ridge density and clarity, high curvature areas, dominant block orientation, intensity, …

Evaluate each block based on features quality (4 – perfect quality … 0 - background)

How big is typically the iris code? Why?

The standard iris code is **2048** bits (plus 2048-bit mask).

128 * 8 - each pixel is 2 bits

When it comes to faces, what is the basic resemblance between how do people recognize faces and how the CNN recognizes them?

Humans perceive the face based on certain **higher-level** characteristics, while the **CNN** perceives **low** and **high-level** characteristics.

Describe the properties of multimodal systems.

- It combines different types of modalities together.
- Traits are expected to be uncorrelated, which leads to greater improvement in performance compared to other types of multibiometric systems.
- Higher cost.
- There are diminishing returns after strong biometric traits have been fused.

2017-00-00

Name and explain briefly each of the authentication types.

Pair matching: Do these images match?

Verification: Does this sample match the one in our system? (1:1)

Identification: Does this sample exist in our system? (1:n)

Negative authentication: I'm not the subject X / I'm not a member of the group X

Deduplication: Do we have to enroll you?

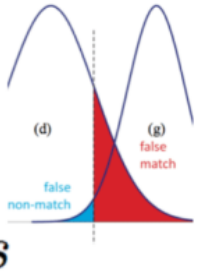
At which stage in the recognition pipeline do we use the similarity score and the distance score? What is the difference between the similarity score and the distance score?

The similarity and distance score are used in the **matching stage** of the recognition pipeline. We want the **similarity** score to be as **high** as possible, but we want the **distance** score to be as **low** as possible.

However, if the distance score is equal to 0, we should rather view it as an impostor, because of the really low probability of that actually happening.

Explain how do we calculate False Non-Match Rate? Also, sketch and mark it using the genuine and impostor distributions.

False Non-Match Rate



- If we know g: $$g_{FNM}(\tau) = \int_{-\infty}^{\tau} g(s)ds$$
- But typically we don't, thus:

$$\hat{g}_{FNM}(\tau) = FNMR(\tau) = \frac{\text{Number of false non-matches for } \tau}{\text{Number of all genuine comparisons}}$$

Explain measures AUC and EER. What does it mean if AUC is 0.5?

AUC is used to **compare** the **performance** of **two systems** and the values **range** from **0.5 to 1**.

EER refers to **that point in a DET/ROC curve** where the **FAR equals the FRR**. A **lower EER value** indicates **better** performance.

If **AUC is 0.5** it is the **worst-case scenario** for the biometric system.

Which factors must be considered to determine the suitability of a physical or a behavioral trait?

Universality: possess the trait – minimize the failure to enroll

Uniqueness: different traits – the FMR should always be low enough

Permanence: invariant over a period of time – if not it may lead to high FNMR

Measurability: possible to acquire and digitize – can we measure something

Performance: how many samples can we process in a period of time

Acceptability: willingness to present the trait (Indians don't accept shaking hands)

Circumvention: imitate traits – evade recognition

Name the fingerprint pattern types that are based on spatial relationship between singular points. To which type the majority of prints belong to? What is the expected number of singular points within a fingerprint?

Types based on special relationship between **one loop and one delta** are: **tented arch, left loop and right loop**. The types based on **two loops and two deltas** are: **whorl and twin loop**.

The **majority** of prints belong to the **left loop**.

Within a fingerprint there are either **no singular point, 2 singular points or 4 singular points**.

How do we extract minutiae from binarized and thinned fingerprint image?

We extract the minutiae by looking at the immediate neighborhood of a pixel. If it has two ridge points than it is just a ridge. If it has **one ridge point**, then it is a minutia (**ridge ending**) and if it has **three ridge points**, then it is a minutia (**ridge bifurcation**).

How many pixels across the iris in the radial direction for successful processing do most iris recognition systems expect? Why do we need to use near-IR spectrum for iris acquisition?

Most systems expect **100 to 200 pixels** for successful processing.

Because using near-IR illumination **allows usable texture imaging even for dark irises**.

In face detection step we used integral image. Why do we need it and how do we calculate it? How many cascades do we need to reach a GMR of 0.9 and a FMR of approximately 10−6, if each cascade has a GMR of 0.99 and a FMR of 0.3?

We need the integral image to **reduce the number of calculations** needed as we need to calculate the sums of "rectangles" in an image. By using the integral image, we **reduce** the number of calculations for any rectangle to **4**. We calculate it by creating a new array where **each pixel** is a **sum of all pixel intensities to the left and above it in the original image**.

Assume each stage has a GMR of 0.99 and FMR of about 0.3:

10 cascades – (**0.99^10 ~= 0.9**) and (**0.3^10 ~= 10^-6**)

((0.99^**X** ~= 0.9 & 0.3^**X** ~= 10^-6))

What is the difference between Eigenfaces and Fisherfaces? Which methods are behind each of them?

Eigenfaces attempt to maximize the scatter of the training images in faces space (**PCA method**)

Fisherfaces attempt to maximize the between class scatter, while minimizing the within class scatter (**LDA method**)

2025-01-30

Name and describe the differences in enrolment and recognition pipelines

Enrollment Pipeline

- New user comes in
- Acquisition ⟹ Feature extraction ⟹ Template generation

Recognition Pipeline

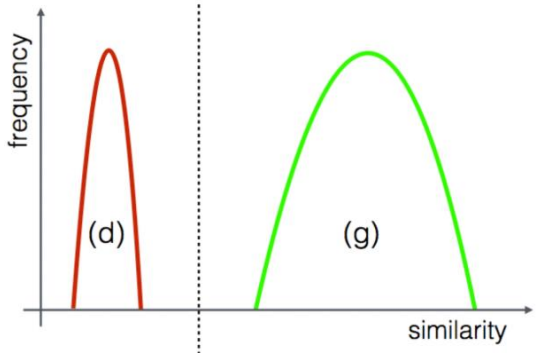
- We need to make a decision
- Acquisition ⟹ Feature extraction ⟹ **Matching ⟹ Labeling**

What is the ideal and the real-world distribution of match scores. Define the problems if there are any.

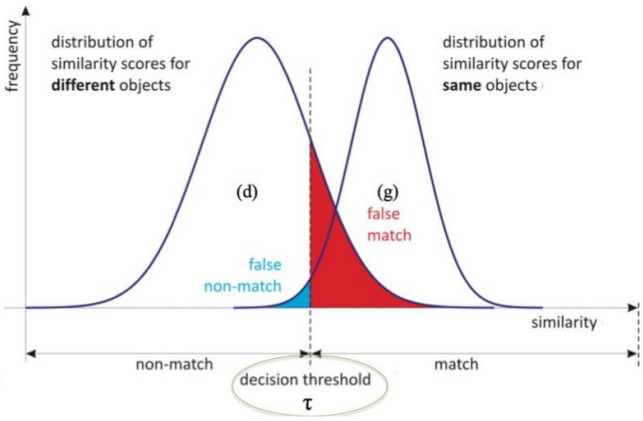
Score distributions (d- different, g- genuine)

Ideal

- Well-separated



Real



Problem is in decision threshold. The FAR and the FRR cannot be reduced simultaneously by adjusting the threshold!

Explain minutiae matching. Use pseudo-code to do so.

Pairing Minutiae

- Minutia *a* is in correspondence with *b* if their **distance** is within a predefined distance threshold (e.g., 15 pixels).
- The angle between their directions is within another predefined **angle** threshold (e.g., 20 degrees).
- *a* is allowed to match to **at most one** *b*, and vice versa.

input : Two minutiae sets $\{x_i^T, y_i^T, \theta_i^T\}_{i=1}^{M}$ and $\{x_j^Q, y_j^Q, \theta_j^Q\}_{j=1}^{N}$;
Transformation parameters $(\Delta\theta, \Delta x, \Delta y)$
output: List of paired minutiae
Initialize: set flag arrays f^T, f^Q, and *count* as 0; *list* as empty
for $i = 1, 2, \cdots, M$ **do**
  for $j = 1, 2, \cdots, N$ **do**
    if $f^T[i] == 0$ & $f^Q[j] == 0$ & *distance between minutiae i and* $j < t_d$
    & *rotation between them* $< t_r$ **then**
      $f^T[i] = 1$
      $f^Q[j] = 1$
      $count = count + 1$
      $list[count] = \{i, j\}$
    end
  end
end
return *list*

What are the levels in face recognition. Name them based on the features used in each level. Approx what IPD inter-pupillary distance) do we require for each level.
L1 – **geometry** of the face and global skin **color** (short round face / elongated thin face, male / female, different races) (**<30 IPD**)

L2 – structure of the face components (eyes), the relationship between facial components, precise shape of the face (**30 to 75 IPD**)

L3 – **micro level** features on the face like scars, freckles, moles, … (**>75 IPD**)

Why do we use a mask in iris recognition. How do we normalize the iris?
We use a mask to **mask out the noise in the image** (eyelids, eyelashes), because we don't want them to **impact** the **feature extraction**.

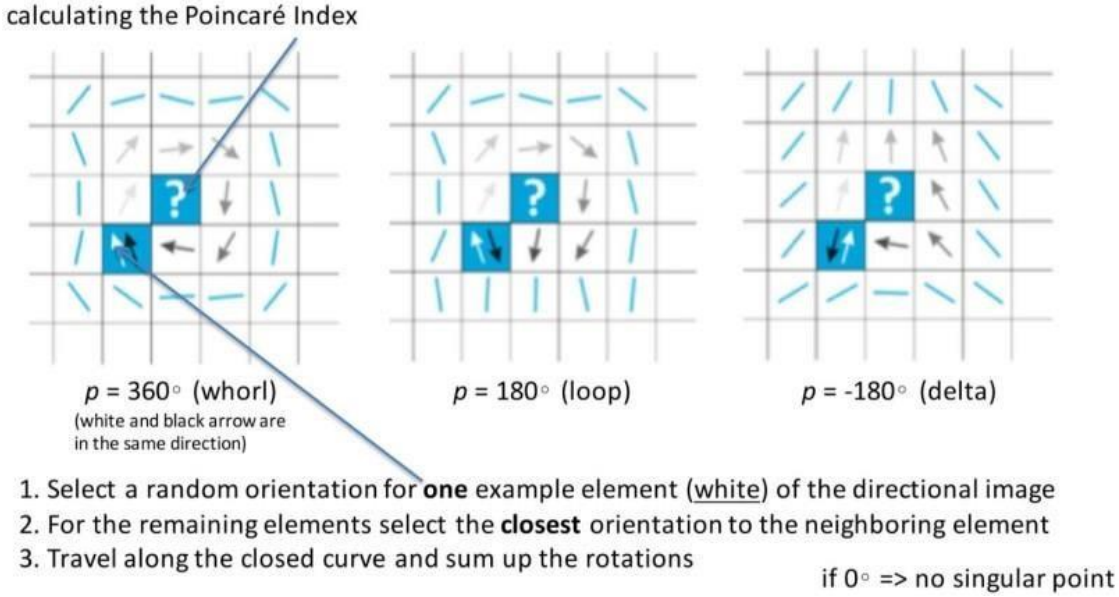
We use the Daugman's **rubber sheet model**, which **is linearly stretching** or **compressing** the imaged iris to **fit a size frame**.

Explain HOG for face detection. How do we use it in matching?
The face image (based on landmarks) is **divided into cells**. For each cell, we compute a **histogram of gradient directions**. Then, histograms of all cells are **concatenated into a single feature vector**. At the end we compare these vectors for a **match**.

What does ROC curve tell us about the system. How do we compare different systems based on it?
Receiver Operating Characteristic Curve. It **simplifies** the **comparison** of **two** systems with **different** thresholds.
If the **FRR** of one biometric system **(A)** is consistently **lower** than the **FRR** of the other system **(B)** for corresponding values of **FAR**, one can conclude that the matching performance of biometric system **A** is better than that of **B.**

What can be seen when observing the Fourier transformation of local points in iris?
- **Out of focus blur**: Fourier spectrum is supposed to be largely dominated by the low frequency components.
- **Motion blur**: Lacks middle and high frequency components and has frequency distribution similar to that of an out-of-focus image.
- **Occlusion due to the eyelashes, eyelids**: Contains significant middle and high frequency components.

Explain Poincare index. Where did we use it?
- Is cumulative change of orientations along a closed path in an orientation field
- Generally evaluated using the eight neighbors of a pixel



It is used for **singularity extraction** in fingerprints.

How are feature pyramids used in RetinaFace?
Feature Pyramid Network (FPN):
- **Bottom-up pathway**: Extracts **high-level semantic features**.
- **Top-down pathway**: Enhances **low-resolution features** with semantic information.
- Ensures effective face detection across **multiple scales**.